Assignment 3: Conduct a Human-Subjects Study of the AI-Assisted Task

MULTIPLE DEADLINES

Part A3-1
Post task links: due 10:00 **am (before class),** Tue, Nov 12, 2026 on
Assignment 3 HIT links
Expected time: 5-12 hours

Part A3-2 due 11:59 pm, Sun Nov 22, 2026 on Canvas
Expected time: 2-4 hours
You may want to start preparing your data analysis pipeline while waiting for the data to come in

---

Task Description

In this assignment, we will build on Assignment 1 and Assignment 2 to create two versions of web interfaces for your designed task, one with AI assistance and one without. You will also be asked to participate in your classmates' studies at the class on Nov 12, and they will participate in yours. After that, you will analyze the collected data to see whether your prediction in Assignment 2 holds.

*You must attend Nov 12 class in person to collect data from your classmates. Otherwise, you are responsible for collecting your own data outside the class.

A3-1 Create and Deploy Interfaces for the Task You Designed in Assignment 2

You will use the study dataset you created in Assignment 2 as the task trials for participants (your classmates) to complete. For interface implementation, you can build on what you had for Assignment 1, or start from scratch. You can follow these steps:

1. Create a study interface for the **baseline condition without AI assistance**. As in Assignment 1, make sure your instructions are clear and participants' submitted data will be correctly recorded. You may want to visit the instruction for Assignment 1 and GSI's summary for Assignment 1 to make sure you have good designs of workflow, instruction, and task interface.
    a. Like Assignment 1, for your web interface, you must have a backend to deploy the task and store the data collected from participants to earn the full points. If you choose to use a survey platform like Qualtrics. you will lose **10 points** (out of 100 total points of this assignment).
2. Set the number of task trials (N) you would like each participant to complete so it would take them **less than 10 minutes** to complete your study, including reading the

instructions. Similar to Assignment 1, your interface should randomly load N trials from your study dataset for each participant, so that not all participants will work on the same trials (otherwise your study results might be biased by the nature of this small set of tasks).

3. In addition to task performance, design and implement at least one **additional measurement** you want to compare the two conditions with (you are welcome to include more). Your submission should also justify why these measurement(s) matter. Some examples:
    - Task performers' subjective experience (e.g. perceived workload, confidence) can often be measured by a questionnaire issued after the study or individual trial (hint: you are allowed to use a survey tool such as Google Forms for this, but it needs to be able to track who submitted what answers)
    - Efficiency or time spent is another important dimension of task completion
    - Table 4 of this survey paper provides some pointers
4. Your interface should include at least one **attention check** to make sure participants are answering your questions carefully. This may be a simple question about the task instructions, or one with an obvious answer. Participants may be terminated if they fail to pass, and they have to start over again. Note that participants should not be treated with unreasonable standards, so make the question either very easy as long as one is paying attention, or be somewhat tolerant, e.g. only terminating if failing multiple checks.
5. Create another version of the study interface **with the AI assistance** you designed and implemented in Assignment 2. You have two options:
    a. *Asynchronous AI output*: you can present the AI output you obtained for each trial in Assignment 2 as a piece of static information. You should consider how to design the UI presentation (e.g., by a number or a visualization) and the workflow (e.g., should people see AI output directly or upon request?)
    b. **[+3 Point]** *Synchronous AI output*: you can also directly integrate the model or API call you used in Assignment 2 into your interface, so that the participant will get the AI output in real time. This option would be especially beneficial if you want the human task performer to interact with the AI. If you choose this option, your interface should log the outputs shown to participants and follow-up interactions, if any, so you can analyze what happened in the study.
        i. If you hosted your model on Colab in Assignment 2, it is possible to build a web app and API with Colab to run the model online. If you choose this option, you would need to run the server when your classmates are working on your study, as it cannot run forever (it has a time limit).

Submission A3-1:

Due 1: Tue, **Nov 12, 2026 at 10:00 am - before in-class deployment session**

- Put the links of the two interfaces in this shared spreadsheet and fill out required information
- Make sure your interfaces (and models if you are doing synchronous AI) are up and running for the whole duration of the class

During the Tuesday class, each student will choose 3 tasks from the Without AI spreadsheet and 3 tasks from the With AI spreadsheet. After finishing these, you will work on the same 6 tasks in the other condition (this will ensure half of your participants saw each condition first to avoid order effect)

As a participant, make sure to put your name in the participant column on the Google Sheet. After you complete both conditions, provide a comment on how you compare your experiences with the two conditions, and what you liked or disliked about having AI assistance.

If bugs or study design issues were identified during the in-class deployment, you may need to update the interfaces and collect a new batch of data outside the class.
You should collect data from **at least 5 participants** for each of the two conditions.

Suggested **Due**: Friday, due 11:59 pm Nov 13, 2026 to finish data collection so you have enough time to analyze the data.

A3-2: Analyze collected data and reflect on your results

1. Provide a description of your data collection, such as sample size, what was collected, how it was collected (all from the class or some outside?), and anything else GSI should know when looking at your results
2. Calculate and compare participants' task performance between the two conditions. Most likely you will be able to use the same performance metric(s) used in Assignment 2.
3. Calculate and compare any additional measurements you collected in the study. Describe what metrics you used.
4. **[+2 Point]** Conduct statistical test(s) on the measurement(s) comparing the two conditions. It is possible that you won't see a significance given the relatively small sample size. You will get the full bonus points as long as you used the right test. This website is a good resource for statistical methods commonly used in HCI.
5. Based on the results, reflect on the following. You may need to delve deeper into your data, look at errors or individual data points, and the qualitative data collected from participants' comments.
    a. By the performance measure, does your prediction in Assignment 2 hold? Why or why not?
        i. A good place to understand the why is to analyze the error cases in different conditions (and AI-only errors in Assignment 2), to see whether some patterns were eliminated, added or changed.
        ii. Hint: with the small sample size, it is also possible that your result is biased by outliers (e.g. someone might have misunderstood the task). If this is the case, you should describe how you made that diagnosis.
    b. For any additional measurement(s) beyond task performance, compare them between the two conditions and reflect on why you see certain results.

Submission A3-2:
Submit to Canvas:

1. A PDF including:
    - Links to your two versions of the interface and a description of your task and the AI assistance, including design decisions you made (if it is impossible for your links to be continuously running, e.g., you are using colab, it is acceptable to include detailed screenshots of your interface in your submission, and an explanation of why your link may not be running. If needed, GSI may contact you to run it within a timeframe.)
    - Report of your data analysis, including a description of data collection, what measurements you used and why you made those choices.
    - Your reflection on the results, following Step 5 in A3-2
    - If you did any of the bonus steps, please mention it and provide a description of what you did.
2. Files of your collected study data, with necessary descriptions for TA to understand what was collected.
3. Relevant code (e.g., if you set up your own interface or integrated a synchronous AI function to get the bonus point) for the task interfaces.
4. Relevant code or files for data analysis. TA should be able to replicate your results based on your description and the files you updated.

Grading Rubrics

The assignment grade will be based on the completeness of the following:

- All submissions/dues of A3 are finished by the deadlines
- Have completed at least 6 tasks of others' in both conditions, and submitted your name and comments on the spreadsheets
- The two task interfaces are running with no critical bugs at the time of grading. Provided descriptions (and supporting code or files) for design decisions and bonus steps completed, if any.
- Have collected data from at least 5 valid completed participants in each condition, and data is successfully recorded.
- Have conducted proper analysis on the collected data and submitted the results report, including explanations (and supporting code or files) of how and why they were done
- Have provided reasonable reflections on the study results